7 (Non)Strategic Production Planning and Ambiguity

Experimental Evidence

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Introduction

This chapter investigates the effect of strategic production planning on the encoding of prosodic phrasing as a grammatically driven means to disambiguate a syntactically ambiguous word sequence.[1] We conducted a production study in German involving two groups of participants. One group was informed about the ambiguity of the target sentence and the prosodic disambiguation possibilities. The other group was not informed. The main question was whether the two groups would produce different intensities of a specific prosodic disambiguation method: prosodic phrasing. We focused on the strength of the prosodic boundary, measured in terms of the *duration of the pause* between two intonational phrases (IPhs). The underlying idea is that production planning (here: phonological planning) can be nonstrategic or strategic. The term *strategic production planning* is closely connected to its use in rhetoric, where *strategy* is defined as an "analysis of a relevant goal-resistance-medium relation" (Knape et al. 153; translated by the authors). We hypothesize that the uninformed participant group (UnInf-Group) will plan production without a strategy, whereas the informed participant group (Inf-Group) will plan production under the control of a strategy. The measure is the intensity of the prosodic boundary that each group produces in their utterances after a disambiguating context. The question of how a strategy does or does not control the production process will be addressed to follow ("Theoretical Considerations, Research Questions, and Hypotheses").

The main research question is whether informed speakers more readily emphasize prosodic cues in comparison to uninformed speakers. We conducted a production experiment in German that investigated the prosodic phrasing of ambiguous word sequences like "CHRISTOPH MALT PAUL NICHT" (Christoph paints Paul not), as illustrated in (1) (see Féry 100).

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(1) 

Verbs such as *malen* "to paint" can be used transitively or intransitively in German. The word sequence in (1) without punctuation or disambiguating context is thus ambiguous between a fragmentary *stripping* construction[2] (cf. Hankamer and Sag) and a regular subject-verb-object (SVO) structure, which is the preferred interpretation (cf. Winkler, "Ellipsis and Prosody"). In (2a), which represents the stripping structure, the verb *malen* is used intransitively and allows the DP following the verb (DP2) to be understood as the subject of the following clause. In (2b), *malen* is transitive, thus taking DP2 as its direct object, which results in a regular SVO structure.

(2a) [$_{CP1}$ Christoph$_{DP1}$ malt]. [$_{CP2}$ Paul$_{DP2}$ nicht].
Christoph paints. Paul not.
Christoph is painting. Paul isn't.
(2b) [$_{CP}$ Christoph$_{DP1}$ malt Paul$_{DP2}$ nicht].
Christoph paints Paul not.
Christoph is not painting Paul.

The two readings of example (1) differ not only in their syntactic structure but also in their prosodic phrasing. The readings are distinguished prosodically by whether or not there is a prosodic boundary, which is signaled by the presence or absence of a pause after the verb *malt* ("paint.3SG"). The example either means (2a), with a pause, or (2b), without a pause. We follow Steedman in the assumption that the clausal structure is mapped onto prosodic structure (see also Fernanda Ferreira on the relationship between syntactic and phonological representations). Therefore, the two clauses of (2a) are mapped onto two separate IPhs in the stripping construction, with a prosodic boundary audible as a pause between the intransitive verb and the fragmentary nominal *Paul*, as shown in (3a). The single SVO clause in (2b) is mapped onto a single IPh and contains no pause after the verb, as shown in (3b).

(3a) [$_{IPh1}$Christoph malt]. [$_{IPh2}$Paul nicht]. [2IPhs]
(3b) [$_{IPh1}$Christoph malt Paul nicht]. [1IPh]

There are different potential prosodic disambiguation methods to distinguish (3a) from (3b). The predominant method is to produce (3a) as two IPhs with a prosodic boundary between them, and (3b) as one IPh without such a prosodic boundary. As a measure of the prosodic boundaries, we analyzed the length of the duration between the verb and the subsequent

DP2. In the ToBI[3] system, the end of an intonational phrase is generally signaled by a prosodic boundary, which is annotated with *break index 4* (cf. Beckman and Elam). We follow Lehiste, who studied acoustic correlates of prosodic boundaries in syntactic ambiguities and showed that duration is the most reliable cue of prosodic disambiguation: "The means [the speakers] use for disambiguation is mainly manipulation of the time dimension" (Lehiste 119). Duration is also the main prosodic cue to distinguish the two meanings in (1), as suggested by Féry. We selected this particular kind of syntactic ambiguity in German. Other possibilities to indicate the location of a prosodic boundary in (3a) are preboundary lengthening and domain-initial strengthening. All of these cues may serve as perceptual signals for a prosodic boundary between two IPhs.

We base our expectations about the different types of production planning (here: phonological planning) in the Inf-Group and the UnInf-Group on these observations. The critical prosodic cue for the stripping analysis is the pause between the two IPhs in (3a) and the lack thereof in the single IPh reading in (3b). We assume that the strategic phonological planning of informed speakers results in stronger prosodic boundaries and, thus, more pronounced pauses in (3a) than the nonstrategic phonological planning of uninformed speakers.

The chapter is structured as follows: in the following section ("Theoretical Considerations, Research Questions, and Hypotheses"), we will provide the theoretical background on the issues of production planning and phonological encoding, discuss previous experimental studies on disambiguation, and present our research questions and hypotheses. In the next section, "Experimental Evidence," we will first present the method of our experiment, followed by a presentation of the results and a discussion of our findings. We conclude with a general discussion.

Theoretical Considerations, Research Questions, and Hypotheses

The Place of Strategy in Phonological Encoding

Our core claim in this subsection is that nonstrategic and strategic phonological encoding differ in that the former is an automatic process, whereas the latter is a controlled process, that is, a process under executive control. As Bock puts it, "*Controlled processing is strategic*. It is therefore constrained in a number of ways that automatic processing is not, but it is at the same time less rigid in its application" (9; emphasis added).

A language production model like that of Levelt (see also Bock) subdivides the process of utterance planning into subprocesses and identifies subcomponents, that is, modules, responsible for the execution of the subprocesses. Language production starts with the intention of the speaker to convey a particular thought to one or more interlocutors. Levelt's corresponding first model component is the *conceptualizer*, which is responsible for message generation and which delivers a preverbal message as output. The preverbal message is fed into the *formulator*, which is responsible for linguistic encoding. The formulator starts by executing the grammatical encoding, which consists of lexical access, functional assignment, and serialization of constituents. The resulting surface structure is then phonologically encoded. The formulator's output is a phonetic plan or inner speech that is submitted to the *articulator*, which transforms the inner speech into overt speech. In addition to this sequential stream of processing, the production system is connected to the speech comprehension system to enable the speaker to monitor her own speech. Monitoring and message generation in the conceptualizer are under executive control; therefore, monitoring is conceived of as part of the conceptualizer in Levelt's model. Grammatical and phonological encoding as well as articulation are automatic processes.

Participants in our experiment do not fully plan their utterances before producing them. When comprehending a written sequence of words like "CHRISTOPH MALT PAUL NICHT," they assign a syntactic structure and a semantic interpretation to it. Once this is done, they know what they intend to utter. Word sequences in our experiment are preceded by biasing contexts, which strongly suggest either the transitive or the intransitive verb interpretation. Someone who understands that Christoph is not painting Paul will intend to utter this message; someone who understands that Christoph is painting whereas Paul is not painting at all will intend to utter that message instead. There is no need for the production system to carry out any further grammatical encoding. What remains to be done for the language production system is the phonological encoding, followed by the articulation of the inner speech. As noted previously, phonological encoding proceeds automatically in the normal case, that is, if speakers are uninformed. We expect uninformed speakers to not even notice the ambiguity of the word sequences they utter, because *the intended message* is not ambiguous. For speakers who are informed about the ambiguity and about the means to avoid it, the circumstances of speaking are not normal. We expect them to intend to avoid the impending ambiguity—as long as they are able to do so.

The distinction between automatic and controlled processing is a common one in cognitive psychology (cf. Shiffrin and Schneider). Automatic processes are fast but rigid, whereas controlled processes are flexible but slow. Controlled processes tap into central working memory capacities and interfere with one another if running in parallel, whereas independent automatic processes can run simultaneously and demand no attention. Automatic processes are usually unavailable to consciousness. Controlled processes may be available to conscious perception, but they still may not be susceptible to manipulations by verbal instruction. Shiffrin and Schneider emphasize the benefits of these two processing modes for a cognitive system. On the one hand, the ability to automatize cognitive processing relieves the central executive control considerably and thus enhances the flexibility of the system as a whole. Automatization by learning equips the organism with highly adaptive, fast, and efficient processing. On the other hand, Shiffrin and Schneider address the value of control for making adjustments to automatized processing due to environmental changes, because "[i]t allows the organism to adjust to changes in the environment that make previously learned activity patterns useless or harmful" (161).

Is the scene thus set for our claim that speakers can switch the mode from automatic to controlled phonological encoding? According to Levelt (22) it is not: "Formulating and articulating are . . . probably largely impenetrable to executive control even when one wishes otherwise." Phonological encoding is part of formulating, and if formulating is impenetrable to executive control, then the central system cannot influence phonological encoding. Whether an automatic process is impenetrable may depend, however, on how the process is initiated. If an automatic process is initiated automatically, it responds to a particular input pattern, and the process as a whole is impenetrable if, in response to the particular input pattern, it mandatorily delivers a particular output. However, an automatic process might gain flexibility once initiation is controlled, a possibility envisaged by Shiffrin and Schneider (151). Because Levelt (20) characterizes automatic processes as "reflex-like," he seems to reject such an option as reflexes are initiated automatically.

We are inclined to object that Levelt underestimates the capability of speakers to take control over phonological encoding. As noted earlier, control over automatized processes seems necessary when the organism faces the need to adjust otherwise adaptive automatic processing. For instance, speakers with a fully developed competence of their mother tongue, like would-be actors or newscasters, undergo speech training with the goal of improving control over their articulation while doing their job—although not necessarily also at home. In our view, speech training intervenes at the stage of phonological encoding. Similarly, singers, of any music genre, articulate word sequences (i.e. lyrics) differently whether speaking or singing. We nevertheless assume that spoken and vocalized word sequences are encoded within the same subsystem, the formulator. Hence, we imply some systematic flexibility of phonological encoding. Moreover, control over phonological encoding may also be a crucial ability for speakers who master more than one language.

Besides these more general considerations, we wonder how self-monitoring of speech production fits with a complete lack of executive control over encoding processes. In his analysis of how the production system is able to avoid linguistic ambiguity, Victor S. Ferreira takes seriously the idea that executive control cannot be exerted anywhere but in the conceptualizer. Basically, linguistic ambiguity can be avoided either automatically during encoding in the formulator or in a controlled way, be it during message generation in the conceptualizer or via monitoring. The addressee of the monitor's output must again be the conceptualizer. The only option left for the production system to respond to the monitor's feedback is, thus, to modify the preverbal message because only this representation is subject to controlled processing. Given such a conception, informed speakers in our experiment should be unable to respond to the verbal instruction regarding ambiguity avoidance, because message generation is not their business. This means that informed speakers should not behave any differently from uninformed speakers, who are expected to automatically avoid ambiguity.

Hartsuiker distinguishes three components of self-monitoring: trouble detection, interruption, and repair.[4] He states that speakers repair lexical, phonological, prosodic, or morphosyntactic errors. According to our view, what Victor S. Ferreira sketches as the response of the production system to the monitor's output should not be considered a repair in this narrow sense. The monitor's feedback, according to Ferreira, seems to lead inevitably to a modification in message generation. Hartsuiker and Kolk (129) propose a variant of Levelt's monitoring model, for which they assume

> that the conceptualizing and grammatical encoding parts of the restart take only a little time, especially for repairs of phonological errors, in which case the conceptualization and grammatical encoding processes will in general have the same, correct representations still available.

The authors obviously envisage the possibility of a shortcut to repair specifically at the level of linguistic encoding at which the trouble arises. Here we recognize an instance in which executive control may direct linguistic encoding via initiation of the process without penetrating its execution. Whereas the automatically triggered automatic process can only mandatorily map a particular input to a particular output, a controlled initiation contributes to the input and can thus impart flexibility to the input-output mapping that is lacking in the usual case. This much control is needed to actively intervene in the encoding process and to initiate a repair.

As we do not see how a verbal instruction like the one employed in our experiment can exert an effect on phonological encoding other than via executive control, the question becomes an empirical one. If we find that informed speakers systematically differ in their phonological encoding from uninformed speakers, we will take this as evidence that phonological encoding is not completely impenetrable.

Previous Experimental Studies on Prosodic Disambiguation

There has been a great deal of research on prosodic disambiguation, especially with respect to the role of prosodic phrasing. The literature can be divided into two main groups: (1) research investigating how informed and mostly professional speakers use prosody to resolve ambiguous structures and (2) research investigating whether naïve and untrained speakers use prosody to disambiguate scripted as well as natural speech.

The most important studies which address the question of whether speakers are able to prosodically distinguish ambiguous structures have been carried out by Lehiste, as well as by Price, Ostendorf, Shattuck-Hufnagel, and Fong for English and by Féry for German. These studies predominantly tested partially informed and professional speakers.

Lehiste comes the closest to an investigation of speaker strategies; actually, she speaks explicitly of "strategies of disambiguation." She compared the different productions of speakers who were first left naïve and were only later informed about the ambiguity of the test items. Her focus was on different types of surface and deep structure ambiguities, like "The hostess greeted the girl with a smile" (surface structure ambiguity with different bracketings for the two readings) or "Visiting relatives can be a nuisance" (deep structure ambiguity with identical bracketing for both readings). In the experiment, four speakers (two linguists and two nonlinguists) were asked to read 15 sentences aloud without being given a context or any further information. Afterwards, the same speakers were informed about the ambiguity of the 15 sentences in the form of paraphrases and were asked to produce each sentence again, once for each meaning, "making a conscious effort to convey one or the other meaning" (Lehiste 107). The productions were then played to 30 listeners (again, half were linguists and half nonlinguists) in a perception study, in which the task was to decide which paraphrase each recording represents. It turned out that the eight surface structure ambiguities were much more successfully resolved by speakers and listeners than the seven deep structure ambiguities, in agreement with an earlier suggestion of Lieberman. Lehiste identified timing as the most reliable means for teasing apart the different structures of surface structure ambiguities: a pause coinciding with a phrase boundary lengthened the articulation of a word sequence in comparison to the same word sequence without a pause and phrase boundary. Professional and untrained speakers or listeners performed about equally well, but Lehiste found that making the speakers aware of the ambiguities led to better performance of speakers and listeners for five of the surface structure ambiguities.

Price et al. conducted a similar study for English with surface structure ambiguities and professional speakers. They found that naïve listeners are able to correctly disambiguate a variety of syntactic ambiguities produced by four speakers. They also found that syntactic boundaries coincide with prosodic boundaries.

In a production experiment, Féry investigated prosodic disambiguation possibilities in German. Speakers had to prosodically distinguish a series of 20 syntactic ambiguities which were presented to them with different syntactic bracketings. Of the five speakers, only three were able to prosodically disambiguate the sentences. Féry analyzed the prosodic cues used in the successful productions to distinguish the two readings. She concluded that German syntactic ambiguities are mostly resolved by durational differences such as pauses, preboundary lengthening, and different types of boundary tones.

In more recent research, the focus has shifted toward the question of whether untrained naïve speakers also make such prosodic distinctions in scripted conversation as well as in natural language production. Allbritton, McKoon, and Ratcliff compared the recordings of professional and untrained speakers who were informed or uninformed with respect to the syntactic ambiguity of the sentences. They found that neither of the uninformed groups produced enough prosodic cues to distinguish the sentences and, accordingly, they argued that the conclusions of previous studies (e.g., Lehiste; Scott; Price et al.) do not hold, in general.

Schafer, Speer, Warren, and White criticized the unnatural settings of previous experiments and introduced a cooperative game task to elicit spontaneous rather than scripted speech. The naïve participants interacted naturally with each other during the game, thereby producing potentially ambiguous structures such as, "I want to change the position of the square with the triangle." The verbal interaction was recorded. Contrary to Allbritton et al.'s findings, Schafer et al. observed that participants consistently produced prosodic cues to disambiguate between the high- and the low-attachment readings of the sentences. They suggested that these different results might be due to the speakers' awareness that they had a communicative task to fulfill.

Snedeker and Trueswell offered an alternative explanation for the disagreement between the findings of Schafer et al. and those of Allbritton et al. They claimed that a speaker's use of prosodic cues depends on how strongly a given context disambiguates a sentence. To show this, they compared the productions of informed and uninformed speakers in a referential game task with an ambiguous and an unambiguous setting. Participants were prompted to naturally produce sentences like "Tap the frog with the flower" for an explicit listener. They found that participants used strong prosodic cues in the form of durational differences only if the situational context did not already disambiguate the sentence toward a high- or low-attachment reading. The results thus provide evidence for Snedeker and Trueswell's claim that speakers only use prosody to disambiguate if the context does not provide enough disambiguating information.

We can thus conclude that the probability that a speaker will successfully use prosody for disambiguation depends on the following factors: first, the type of ambiguity, for example, deep structure vs. surface structure ambiguities (Lehiste); second, the (un)informedness of the speakers (Lehiste; Allbritton et al.); third, a speaker's consciousness of the communicative goal (Schafer et al.); and fourth, the type of context, that is, whether it disambiguates an utterance and triggers only one reading (Snedeker and Trueswell).

Although extensive research has been done on the questions of when and why speakers make use of prosodic cues to disambiguate a sentence, not much is known about whether speakers make use of certain prosodic strategies. We argue that speakers do deliberately use prosodic cues in read speech if the following factors are considered: first, the type of ambiguity must be a surface structure ambiguity (as suggested by Lehiste). Second, participants should be placed in a communicative situation wherein they plan their utterance for a real or an imaginary listener (as suggested by Schafer et al.). With regard to informedness, Lehiste showed that informed speakers perform better than uninformed speakers at disambiguating surface structure ambiguities. However, Allbritton et al. found that naïve speakers do not produce strong enough prosodic cues to disambiguate a variety of syntactically ambiguous sentences embedded in 2- to 6-sentence-long contexts in read speech. Snedeker and Trueswell argued that the reason could be too much disambiguating information in the context sequences, which makes further disambiguation through prosodic cues redundant. Remmele, however, found that native speakers of English use pitch accents in order to emphasize the antecedent of a globally or temporarily ambiguous sluicing structure, despite the presence of disambiguating information in the form of a preceding context or morphology.

In contrast to these findings, we argue that informed as well as uninformed speakers do in fact use prosodic cues to distinguish the two readings of the word sequence illustrated in (1), even if enough disambiguating information is provided by the context. We base this prediction on the assumption that the clausal structure is mapped onto the prosodic structure of a sentence. Therefore, the two clauses of the stripping construction [cf. (2a)] are mapped onto two separate intonational phrases with a prosodic boundary between the verb and DP2. The prosodic boundary is realized by a pause of variable duration. Note that Lehiste did not compare durational differences between the productions of informed and uninformed speakers.

Research Questions and Hypotheses

Two research questions arise regarding prosodic disambiguation and how it relates to nonstrategic vs. strategic phonological encoding. The first question is whether uninformed speakers automatically disambiguate the word sequences in question by means of prosodic phrasing, even if a strongly biasing context may leave the impending ambiguity unnoticed. If so, the second question is whether informed speakers show strategic control of prosodic encoding in response to being informed by taking considerably more effort to encode the prosodic boundary in 2IPh utterances in comparison to 1IPh utterances without a pause. We formulate hypothesis H1 to address the first research question and H2 to address the second research question.

(4) Hypotheses

H1 (UnInf-Group): Uninformed speakers automatically disambiguate word sequences by prosodic phrasing, in that they produce one intonation phrase for 1IPh utterances and two intonation phrases for 2IPh utterances:

The time from the offset of the verb to the onset of DP2 is longer for 2IPh utterances than for 1IPh utterances (*nonstrategic phonological encoding*).

H2 (Inf-Group): Informed speakers make more of an effort to mark the prosodic boundary between the two intonation phrases of a 2IPh utterance, that is, informed speakers produce more pronounced prosodic boundaries than uninformed speakers:

The difference in time from the offset of the verb to the onset of DP2 between 1IPh and 2IPh utterances is larger for informed than for uninformed speakers (*strategic phonological encoding*).

Experimental Evidence: Production Study

To test our hypotheses H1 and H2, we designed a production experiment in which native speakers of German vocalized ambiguous sequences of words. Two groups were tested: an informed group of participants who were made aware of the ambiguities (Inf-Group) and an uninformed group of participants who were not made aware of them (UnInf-Group). The goal was to investigate (1) whether there is a prosodic difference between the two alternative readings of the ambiguous word sequences and (2) whether this prosodic difference is more pronounced in the Inf-Group than in the UnInf-Group.

Method

Design

The experiment implemented a 2×2 factorial design in which the within subjects factor *IPh-Number* (1IPh vs. 2IPh) was crossed with the between subjects factor *Group*[5] (UnInf-Group vs. Inf-Group).

Participants

Twenty-one native speakers of German took part in the experiment, all of whom were BA, MA, or PhD students at the University of Tübingen. One participant was excluded because s/he misunderstood the experimental task. Ten participants each were randomly assigned to the UnInf-Group and to the Inf-Group. Of the participants in the UnInf-Group, one was male and nine were female; they were between 20 and 27 years old, with a mean age of 23 years. Of the participants in the Inf-Group, two were male and eight were female; they were between 22 and 42 years old, with a mean age of 28 years. Participants in the UnInf-Group were rewarded with €10, and those in the Inf-Group were given €15.

Material

The material consisted of 12 experimental items and 14 filler items, all of which were ambiguous between two readings. The list of experimental items is provided in the Appendix. An example of an experimental item as presented on single cards to the participants is illustrated in (5). (5) can have the two meanings in (5a) and (5b), in which punctuation is used to disambiguate the sequences.[6]

(5)

(5a) [$_{CP1}$ Janina$_{DP1}$ badet]. [$_{CP2}$ Nadine$_{DP2}$ nicht]. 2IPh condition
Janina bathes. Nadine not.
Janina is bathing. Nadine isn't [~~bathing~~].

(5b) [$_{CP}$ Janina$_{DP1}$ badet Nadine$_{DP2}$ nicht]. 1IPh condition
Janina bathes Nadine not.
Janina is not bathing Nadine.

Each item was preceded by a short explanation of context that disambiguated the word sequence toward the 1IPh or the 2IPh reading. The contexts consisted of three sentences. The first sentence, S1, introduced the scene setting; the second sentence, S2, introduced the referents mentioned in the target word sequence; and the third sentence, S3, set a strong bias toward the intended reading. The two contexts for (5) are given in (6). (6a) sets the bias for condition 1IPh, and (6b) sets the bias for condition 2IPh.

(6a) S1 *Set Scene:* Kleinkinder brauchen noch viel Hilfestellung bei
Small children need still much help with
alltäglichen Dingen.
daily things.
Small children need a lot of help with daily activities.

S2 *Introduce Referents*: So ist die kleine Nadine beim Baden noch auf die
So is the little Nadine when bathing still on the
Unterstützung ihrer Mutter angewiesen.
support of her mother dependent.
That's why little Nadine is dependent on help from her mother when she is taking a bath.

S3 *Create Contrast:* Aber Janina hat heute leider kaum Zeit und
But Janina has today unfortunately hardly time and
überlegt, wo sie Abstriche machen kann.
thinks, where she deductions make can.
Unfortunately, Janina is in a hurry today and is thinking about where she can save time.

Target: (5) JANINA BADET NADINE NICHT

(6b) S1 *Scene Setting:* Am Strand von Mallorca ist immer viel los.
At the beach of Mallorca is always much going on.
There is always a lot going on on the beach of Mallorca.

S2 *Introduce Referents*: Janina und Nadine liegen in der Sonne und genießen
Janina and Nadine lie in the sun and enjoy
ihren Sommerurlaub.
their summer vacation.
Janina and Nadine are basking in the sun and enjoying their summer vacation.

S3 *Create Contrast:* Während Janina ab und zu ins Meer springt,
While Janina now and then into the sea jumps,
hat Nadine Angst vor Haien.
has Nadine fear of sharks.
While Janina jumps into the water now and then, Nadine is afraid of sharks.

Target: (5) JANINA BADET NADINE NICHT

The two versions of the items, 1IPh vs. 2IPh, were assigned to two different lists according to a Latin square design; both lists were tested by an equal number of the participants in each group. Hence, each participant produced every item in only one of the two conditions of IPh-Number, but every item was tested equally often in either condition.

The 12 experimental items were subjected to a written pretest to ensure that the contexts are strong enough to trigger only one of the two readings. Four independent, naïve participants read all context-target pairs in the same randomized order and added punctuation marks to the target word sequences, which were presented to them in capital letters. The punctuation added by the four participants agreed with the intended readings without exception. When asked about the difficulty of the task, they all indicated that they encountered no problems whatsoever and that they considered the contexts to be unambiguous.

The 14 filler items consisted of different types of ambiguities, some of which can be distinguished prosodically, such as *secondary predicates* (7), *homographs* (8), *attachment ambiguities* (9), and *focus ambiguities* (10). The filler items served to keep participants from getting used to the specific structure of our experimental items, which might have resulted in less informative productions. Although some of the ambiguities of the filler items suggest prosodic disambiguation, we focus on one specific structure in this chapter, namely, the stripping vs. SVO ambiguity exemplified in (6).

(7) Max kaufte den Laden leer.
Max bought the store empty.
Max bought everything that the store offered for sale. vs. Max bought the store in an empty condition.

(8) Der Kapitän muss übersetzen.
The captain has to translate/cross over.
The captain has to translate. vs. The captain has to cross over.

(9) Der Polizist verfolgte den Dieb mit dem Fahrrad.
The police officer chased the thief with the bicycle.
The police officer chased the thief who was on the bicycle. vs. The police officer, who was on the bicycle, chased the thief.

(10) Anna mag nur grüne Tomaten.
Anna likes only green tomatoes.
Anna only likes green tomatoes (not red ones). vs. Anna only likes green tomatoes (not other green fruits).

Procedure

Participants were tested individually in a quiet study room equipped with a stereo microphone with 96-kHz/24-bit recording; the full experimental sessions were recorded. Instructions and materials were presented to participants by means of Power Point slides. Each presentation started with a set of instructions that was dependent on the group, followed by an exemplary item that showed the context and the target word sequence, as well as when to vocalize the target word sequence.

UnInf-Group: The instructions presented to participants of the UnInf-Group explained the task step by step without telling them anything about the stimuli. On each trial, participants were presented with one of the two disambiguating contexts and were asked to vocalize the target sequence as a continuation of the context.

Inf-Group: The instructions presented to participants of the Inf-Group provided them with further information in addition to the task description. Firstly, they saw both context versions, with the version for subsequent vocalization visually highlighted.[7] Secondly, the instructions pointed out that the target word sequences are ambiguous and that the two different contexts disambiguate the sequences toward the two different readings. Thirdly, they listened to a sample recording and were shown some general possibilities of how to prosodically distinguish different readings of a sequence of words; the sample sequences did not, however, correspond to the ambiguities used in the experiment. Fourthly, the instructions emphasized the communicative character of the task: participants should aim for a distinct prosodic differentiation to enable hearers in a subsequent perception study to identify the corresponding context.

The instructions were followed by a short practice session with three items, and then the experimental session started. Up until the end of the practice session, the experimenter was present to answer any questions the participant might have. With the beginning of the actual experiment, the participant was left alone to ensure that s/he felt comfortable while reading aloud and was not influenced by the experimenter's presence. The communicative goal was pointed out by mentioning a subsequent perception task (Inf-Group) as well as by having the participants produce their sentences as an answer to the question, "What happened?" (Inf-Group and UnInf-Group). The broad focus question "What happened?" was chosen to keep the articulations as unaffected by information structure as possible (cf. Wagner; Remmele).

Experimental and filler items were intermixed with one another and presented to participants in a single randomized order. Participants clicked through the slides at their own pace, that is, they took as much time as they needed to complete the experiment. Trials were handled one after another and without jumping back to an earlier trial. On average, participants of the UnInf-Group and the Inf-Group took about 30 and 45 minutes, respectively. On each trial, participants first saw one of the two (UnInf-Group) or both (Inf-Group) contexts together with the target word sequence in capital letters and were asked to read through the material carefully (slide 1 for both groups). One of the two contexts presented to the Inf-Group was then visually highlighted, indicating the context to be considered for subsequent vocalization (slide 1b for the Inf-Group). Finally, participants were prompted to read aloud the target word sequence, keeping the respective context in mind (slide 2 for both groups).

Results

Data Preparation

The 10 participants in each group vocalized 12 target word sequences: six in condition 1IPh and six in condition 2IPh. There were, thus, five vocalizations per experimental item per context per group and 240 vocalizations altogether. After having manually extracted these 240 vocalizations from the recordings, we analyzed them with the acoustic analysis software *Praat* (Boersma and Weenink). We conducted time measurements at specific points in time with the help of a *Praat* script, as illustrated in (11).

(11) /Stef/fi//gra/tu/**liert//Mar**/ti/na//nicht/

For our present purposes, we were interested in the duration of the interval from the offset of the last syllable of the verb (**liert**) to the onset of the first syllable of the DP2 (**Mar**), henceforth called the *critical duration*. We analyzed whether these critical durations differ between the 1IPh and the 2IPh condition.

Analysis of Prosodic Phrasing

We computed ANOVAs with participants (F_1) and items (F_2) as random factors. *IPh-Number* (1IPh vs. 2IPh) and *Group* (UnInf-Group vs. Inf-Group) were used as fixed factors. Figure 7.1 shows the mean values of the critical duration in the four conditions.

The analyses yielded a significant effect of *IPh-Number* on the critical duration [F_1 (1,18) = 67.7, $p < 0.001$; F_2 (1,11) = 183.7, $p < 0.001$]: critical durations were, on average, longer in the 2IPh condition than in the 1IPh condition. Moreover, *IPh-Number* interacted with *Group* [F_1 (1,18) = 7.2, $p < 0.05$; F_2 (1,11) = 16.3, $p < 0.01$]. The difference in the critical duration between the 2IPh and the 1IPh condition is larger in the Inf-Group compared to the UnInf-Group, in agreement with hypothesis H2. We computed a paired *t*-test for the UnInf-Group to confirm that this smaller difference is

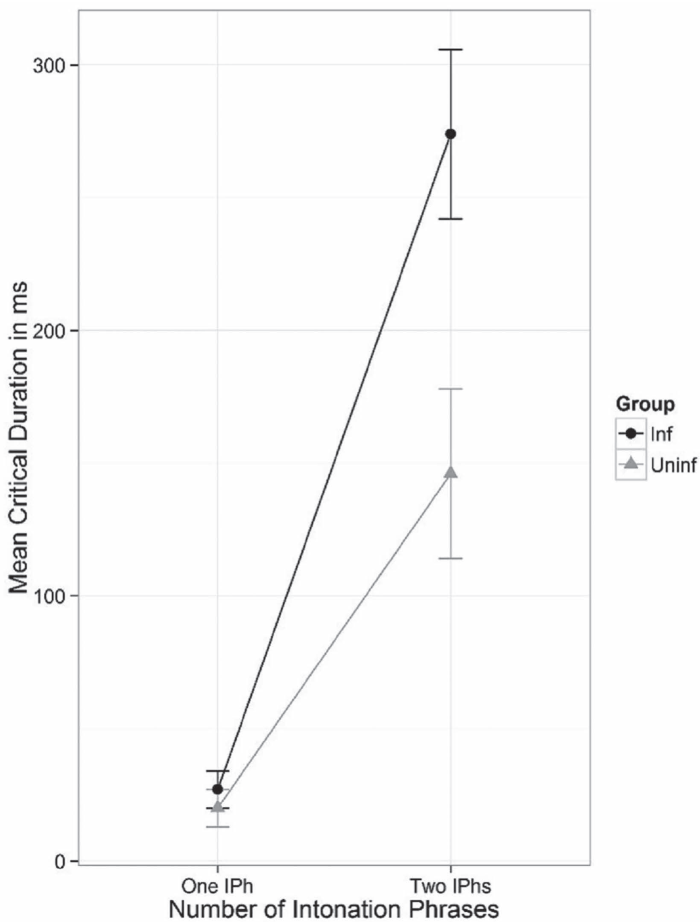

Figure 7.1 Mean critical duration as a function of IPh-Number and Group

significant, as predicted by hypothesis H1. This smaller difference turned out to be reliable [UnInf-Group: $t_1(9) = 3.90$, $p < 0.01$; $t_2(11) = 6.48$, $p < 0.001$].

To summarize, both hypotheses were supported by the analyses. Stripping constructions were prosodically distinguished from SVO structures with the help of prosodic phrasing, that is, an intonational phrase boundary between the verb and DP2 (a 4 on the break index tier). As is evident from the interaction, the Inf-Group made an extra effort to increase the prosodic differences in the form of a more pronounced pause in condition 2IPh. Whereas the mean difference between the critical duration in condition 2IPh vs. 1IPh was 126 ms in the UnInf-Group (2IPh = 146 ms minus

1IPh = 20 ms), it was about twice as large, 247 ms, in the Inf-Group (2IPh = 274 ms minus 1IPh = 27 ms).

Analysis of Intonation Contours

An analysis of two exemplary pitch extraction contours of the Inf-Group further illustrates the results of the production experiment. A vocalization of the stripping construction reading (2IPh) by an informed speaker is visualized in Figure 7.2 [(cf. (2a)]. The prosodic realization contains a break index 4 after the verb and thereby signals the division into two IPhs, where the second IPh is an instance of stripping in German. The prosodic realization follows from the *parallelism condition* (cf. Carlson), which predicts that parallel analyses of conjoined structures are preferred over nonparallel ones (here *parallel* means that similar constituents end up in similar syntactic roles): the subjects *Steffi* and *Martina* are realized with rising pitch accents (L+H*) and are therefore interpreted as contrastive topics. The verb *gratuliert* "congratulate.3SG" in the first conjunct is assigned a focus accent (H*), as is the negative particle *nicht* (not) in the second conjunct. The parallel prosody supports the interpretation of the two IPhs as two coordinate clauses, where the intransitive verb *gratulieren* (to congratulate) is reconstructed in the second conjunct. The recovery of the ellipsis site in (2a) is dependent on the *givenness marking hypothesis* and the *contrastive remnant condition*.[8] In addition, it conforms to the prosodic requirement of the parallelism condition (cf. Carlson). A vocalization of the SVO reading

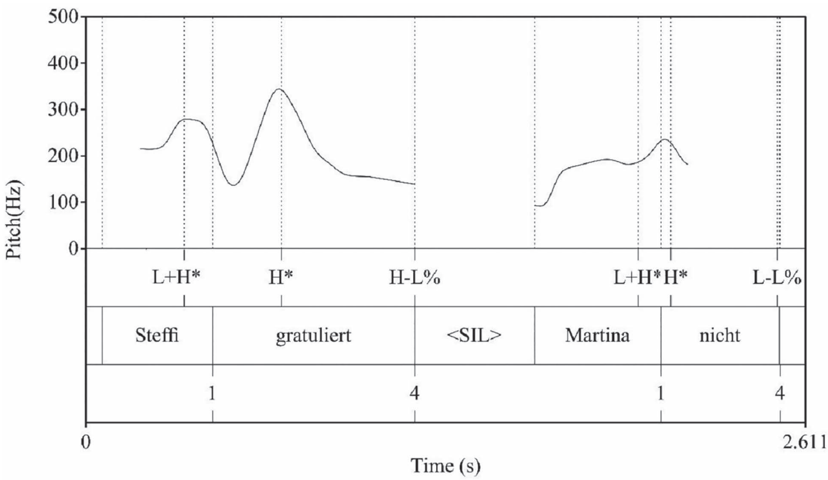

Figure 7.2 Intonation contour of condition 2IPh (Inf-Group)

(1IPh) by an informed speaker is visualized in Figure 7.3 [cf. (2b)]. Here, we have neither parallel pitch assignment with contrastive pitch accents on the DPs nor a break index 4 between the verb and DP2. The parallel prosodic realization of the two IPhs in (2a) supports the recovery of the ellipsis site. From a psycholinguistic perspective, one could argue that there is a temporary ambiguity at the point where the verb *gratulieren* is parsed.

A vocalization of the stripping construction reading (2IPh) by an uninformed speaker is visualized in Figure 7.4. A comparison of Figures 7.2 and 7.4 illustrates the differences in critical duration between the Inf-Group and the UnInf-Group in condition 2IPh. In Figure 7.2 (Inf-Group), we see that the pause between *gratuliert* "congratulate.3SG" and *Martina* is much longer than that in Figure 7.4 (UnInf-Group). In contrast to Figure 7.2, the DPs *Steffi* and *Martina* do not carry contrastive pitch accents in Figure 7.4.

Discussion

The results of this experiment corroborate that informed and uninformed speakers disambiguate word sequences of the type "STEFFI GRATULIERT MARTINA NICHT" by prosodic phrasing. Both the uninformed participants of the UnInf-Group and the informed participants of the Inf-Group prosodically distinguished the two readings of "STEFFI GRATULIERT MARTINA NICHT" by producing a longer duration between the offset of the verb and the onset of DP2 in the 2IPh condition as compared to the 1IPh condition. The longer duration in condition 2IPh signals a pause and thus a phrase boundary, while the very short duration of condition 1IPh

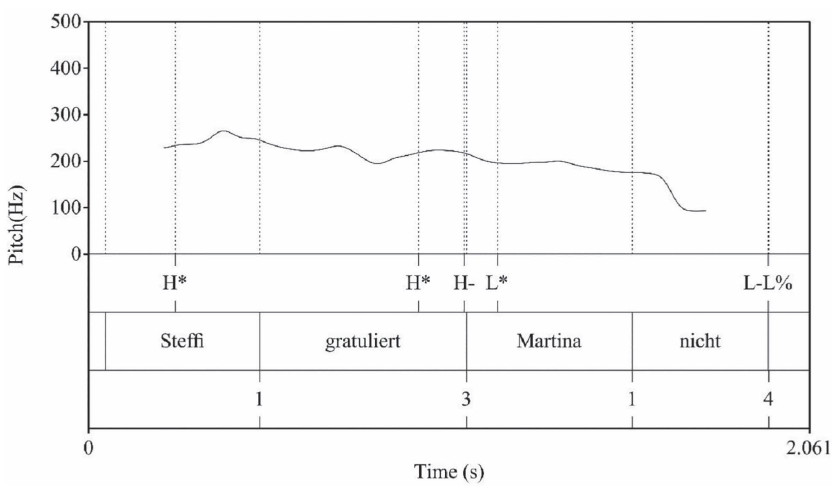

Figure 7.3 Intonation contour of condition 1IPh (Inf-Group)

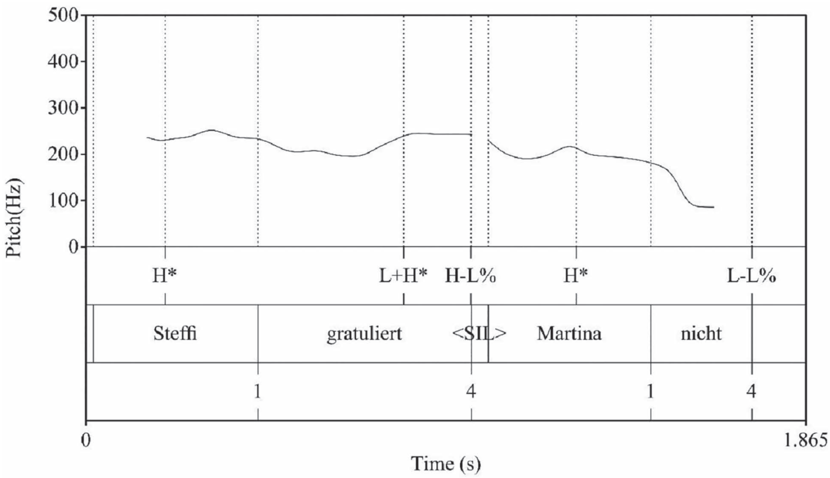

Figure 7.4 Intonation contour of 2IPh condition (UnInf-Group)

signals the absence of a pause and no phrase boundary. This difference between the two IPh conditions is illustrated in Figure 7.1. It indicates that uninformed speakers automatically disambiguate the word sequence by means of prosodic phrasing despite the presence of a strongly disambiguating context. This is the first main result, which confirms hypothesis H1. Secondly, informed speakers distinguish the two readings with extra effort, that is, a 247-ms difference between 1IPh and 2IPh compared to a difference of 126 ms by uninformed speakers, thus confirming hypothesis H2. As the absence of a phrase boundary is evidenced by a very short critical duration in both groups (20 and 27 ms), the difference in prosodic phrasing between the groups is to be attributed to the difference in the length of the pause signaling the phrase boundary present in the 2IPh condition: 274 vs. 146 ms. We take this as clear evidence that the additional instructions given to participants in the Inf-Group are responsible for the substantial group difference. If speakers are made aware of the ambiguous character of the items as well as the fact that the two readings can be prosodically disambiguated, they pay closer attention to their prosodic cues than if they are not informed. Moreover, the Inf-Group was told to imagine that there will be a follow-up group whose task it will be to match their productions back to the respective contexts. With this setting in mind, participants of the Inf-Group paid even more attention to using specific prosodic cues in order to disambiguate the word sequences as clearly as possible. They made use of certain prosodic cues in order to convey a specific meaning and to trigger a specific interpretation by their (implied) hearer. The highly significant effect of the interaction between *IPh-Number* and *Group*, as illustrated in

Figure 7.1, shows that the difference between the UnInf-Group and the Inf-Group is systematic and indicates that the Inf-Group made use of a specific production planning strategy (in this case, a longer duration of the interval between the verb and DP2) in order to set the 2IPh readings apart from the 1IPh readings.

General Discussion and Conclusion

Two central questions have been discussed in this chapter. First, do naïve native speakers differentiate the syntactic ambiguity between SVO vs. stripping by prosodic phrasing? More precisely, do uninformed speakers produce one intonational phrase for SVO structures, but two intonational phrases for the fragmentary stripping construction? We found clear evidence that naïve speakers do exactly this, which supports hypothesis H1. Naïve speakers encoded the stripping construction with a pause of about 150 ms to mark a prosodic boundary between the end of the first intonation phrase, the offset of the verb, and the beginning of the second intonation phrase, the onset of the DP2. As expected, they produced no such pause within the expected single intonational phrase, which corresponds to the SVO structure. In agreement with assumptions by, for instance, Levelt and Bock, we posit that phonological encoding by naïve speakers takes place automatically.

Second, do native speakers who are told to expect ambiguities encode prosodic phrasing differently from naïve native speakers? To put this more precisely: do informed speakers in our experiment make more of an effort to signal the prosodic boundary to their imaginary interlocutors? Again, the answer is yes, which supports hypothesis H2. Informed speakers produced a considerably longer pause to mark the prosodic boundary in the stripping construction than uninformed speakers. We conceive of this lengthening of the pause as an attempt by informed speakers to maximize the probability that the hearer will perceive the prosodic boundary and assign meaning to it. With this conception, we imply that informed speakers pursue a secondary communicative goal besides the primary goal of conveying the intended message to the hearer, namely, the goal of encoding the prosodic boundary in such a way that it cannot be left unnoticed by the hearer. To achieve this goal, the informed speakers in our experiment implemented the strategy of lengthening the pause between the verb and the DP2.

In our section "The Place of Strategy in Phonological Encoding," we argued that prosodic phrasing proceeds automatically when speakers are uninformed, but is controlled when speakers are informed. The question that occurs is whether we should also assume that informed speakers disambiguate the target word sequences by means of prosodic phrasing in an automatic fashion, because we imply that uninformed speakers do exactly this. An automatic process is fast but rigid. It maps a particular input onto a particular output without leaving much room for flexibility in the output. However, in our view, the significantly lengthened pause produced by informed speakers shows such a high degree of systematic flexibility that it seems improbable to assume the same automatic process for uninformed and informed speakers. We do not want to assume two different automatic processes either, with one marking a strong phrase boundary and another one marking a less strong phrase boundary. We argued earlier that automatic phonological encoding is not as impenetrable to executive control as envisaged, for instance, by Levelt, and we have tagged this question as an empirical one. With this in mind, we conclude that the phonological encoding of prosodic phrasing was automatic if speakers were uninformed but controlled if speakers were informed. The most natural place for executive control to intervene in phonological encoding is at the input side: it is the initiation of phonological encoding which is set under executive control and thus enables a systematic modification of the resulting output.

Informed speakers differed from uninformed speakers by the instructions with which they were presented. The additional information given to informed speakers as instructions concerned two issues known to be able to affect phonological encoding in production (cf. previous section, "Previous Experimental Studies on Prosodic Disambiguation"): (1) awareness of an ambiguity and knowledge about prosodic means to differentiate the corresponding readings and (2) susceptibility to the primary communicative goal, that is, the requirement to transmit a message to an interlocutor. We attribute the informed speakers' lengthening of prosodic boundaries first and foremost to the information about the ambiguities and about how to avoid them by prosody. We cannot rule out, however, that the lengthening applied by informed speakers is in part attributable to the emphasis put on the primary communicative goal. More importantly, we cannot assess whether an emphasis on the primary communicative goal would have pushed otherwise uninformed speakers to lengthen the boundary between the two intonational phrases. A teasing apart of the two kinds of information could be easily achieved by providing participants with only one kind of information in a follow-up experiment. In the current experiment, we aimed primarily for a strong effect of the difference in instructions.

According to our interpretation, informed speakers pursue a secondary communicative goal of accentuating the prosodic boundary so that it will not be left unnoticed by hearers. On the basis of our measurements, we clearly see that the pauses between the first and the second intonation phrase are much longer when produced by informed than by uninformed speakers. What we do not see in our measurements is whether the shorter pauses encoded by uninformed speakers run the risk of being missed by hearers or, rather, whether informed speakers unnecessarily exaggerate phrase boundaries. In our view, it would be especially interesting to run an auditory perception study in which hearers interpret the 1IPh and 2IPh vocalizations recorded in the present study to assess the additional value of the lengthened pauses produced by informed speakers compared to uninformed speakers. Because we assume that the vocalizations of informed and uninformed speakers differ in more respects than just pause length, such a study could also deliver some initial indications of which further prosodic cues help hearers to arrive at the intended reading.

We have repeatedly hinted at the rather narrow focus of the current study. We not only concentrated on prosodic phrasing as one possibility of prosodic disambiguation but also paid primary attention to a durational parameter of marking prosodic boundaries. It is therefore desirable to extend the current work to other prosodic means of disambiguation and, not unrelated to this, to investigate additional types of ambiguity. An ambiguity that appears especially attractive to us is a focus ambiguity as was used in the fillers in the present experiment, like *Anna mag nur grüne Tomaten* (Anna likes only green tomatoes). The word sequence can either mean that Anna likes green tomatoes but no tomatoes of a different color, or it can mean that Anna likes green tomatoes but no other green fruits. Depending on the context, either the adjective *grüne* (green) or the noun *Tomaten* (tomatoes) is contrastively focused upon. We expect naïve native speakers to disambiguate the written word sequence by assigning a pitch accent to the contrastively focused expression: the adjective or the noun. It would be interesting to determine whether informed speakers produce heavier accents than uninformed speakers. If so, our current finding would generalize beyond the ambiguity and the prosodic disambiguation cue used here. In particular, a focus ambiguity is not a structural surface ambiguity.

With these future enterprises in mind, we are again committed to the flexibility of speakers in encoding the disambiguation of phonological cues. This flexibility is meant to indicate that phonological encoding does not completely evade executive control; hence, speakers can apply strategies if the circumstances call for more than just fulfilling the primary communicative goal.

Appendix: Experimental Items

(1) CHRISTOPH MALT PAUL NICHT
Christoph paint Paul not

(a) Im Kindergarten ist heute Aktionstag. Betreuerin Susi will zusammen mit den Kindern Bilder malen. Christoph ist gerne kreativ, aber Paul spielt lieber mit Autos.
Today is activity day at the nursery school. Teacher Susi wants to paint pictures with the kids. Christoph likes to be creative, but Paul prefers to play with cars.
Christoph is painting, Paul is not.

(b) Immer freitags findet in der VHS ein Porträt-Malkurs statt. Christoph und Paul sollen sich gegenseitig malen. Aber Christoph hätte lieber einen weiblichen Partner gehabt.
Every Friday, there is a portrait painting class taking place at the adult education center. Christoph and Paul are supposed to paint each other. However, Christoph would have preferred to have a female partner.
Christoph is not painting Paul.

(2) JANINA BADET NADINE NICHT
Janina bathes Nadine not
(a) Am Strand von Mallorca ist immer viel los. Janina und Nadine liegen in der Sonne und genießen ihren Sommerurlaub. Während Janina ab und zu ins Meer springt, hat Nadine Angst vor Haien.
At Mallorca Beach, there is always something going on. Janina and Nadine are lying in the sun, enjoying their summer holidays. While Janina sometimes jumps into the sea, Nadine is afraid of sharks.
Janina is bathing, Nadine is not.
(b) Kleinkinder brauchen noch viel Hilfestellung bei alltäglichen Dingen. So ist die kleine Nadine beim Baden noch auf die Unterstützung ihrer Mutter angewiesen. Aber Janina hat heute leider kaum Zeit und überlegt, wo sie Abstriche machen kann.
Small children still need a lot of help with daily things. This is why little Nadine still relies on the support of her mother when bathing. But Janina hardly has time today and thinks about where to save some time.
Janina is not bathing Nadine.

(3) ANNETTE LOBT CHRISTINA NICHT
Annette praises Christina not
(a) Bei Maren steht wie jedes Jahr ein wunderschöner Weihnachtsbaum im Wohnzimmer. Annette und Christina haben nur eine kleine Tanne in ihrer WG. Während Annette sich an dem schönen Baum erfreut, kann Christina ihren Neid kaum zurückhalten.
As every year, a beautiful Christmas tree can be found in Maren's living room. Annette and Christina only have a little fir tree in their shared flat. While Annette is delighted by the beautiful tree, Christina is hardly able to withhold her envy.
Annette is praising, Christina is not.
(b) Die Musikschule lädt zum alljährlichen Sommerkonzert ein. Christina hat ihre beste Freundin Annette zu ihrem Auftritt eingeladen. Annette ist jedoch alles andere als begeistert von Christinas schiefem Geigensolo.

The music school is holding its annual summer concert. Christina invited her best friend Annette to her performance. However, Annette is anything but excited by Christina's off-key violin solo.
Annette is not praising Christina.

(4) LOUIS ANTWORTET BENNY NICHT
Louis answers Benny not

(a) Herr Schubel hat die Nachbarskinder Louis und Benny beim Grasrauchen erwischt. Natürlich will er wissen, wer den beiden die Droge verkauft hat. Benny stellt sich taub, aber Louis gesteht unter Tränen.
Mister Schubel has caught the neighbors' children Louis and Benny smoking weed. Of course, he wants to know who sold this drug to them. Benny is acting deaf, but Louis confesses in tears.
Louis answers, Benny doesn't.

(b) Manuela hat zwei Söhne im Teenageralter. Louis ist älter und hat daher mehr Lebenserfahrung als der jüngere Benny. Als Benny mehr über Louis' ersten Kuss erfahren will, wird dieser ganz rot und versucht das Thema zu umgehen.
Manuela has two teenage sons. Louis is older and therefore has more life experience than younger Benny. When Benny wants to know more about Louis' first kiss, Louis turns red and tries to evade the topic.
Louis does not answer Benny.

(5) SEBASTIAN GEHORCHT ALEX NICHT
Sebastian obeys Alex not

(a) Stabsoffizier Mayer ist bekannt dafür, besonders rigoros zu sein. Die zwei Soldaten Sebastian und Alex treiben immer gerne Schabernack. Aber bei Herrn Mayer wird Sebastian ehrfürchtig, ganz im Gegensatz zu Alex.
Field officer Mayer is known for being especially rigorous. The two officers Sebastian and Alex like to fool around. However, with Mr. Mayer, Sebastian turns respectful, in contrast to Alex.
Sebastian is obeying, Alex is not.

(b) Herr und Frau Braun sind heute Abend in der Oper. Um den kleinen Sebastian kümmert sich der Nachbarsjunge Alex. Aber Sebastian hat keine Lust, sich an die Regeln des Babysitters zu halten.
Mr. and Ms. Braun are at the opera tonight. The neighbor's boy Alex is taking care of little Sebastian. But Sebastian does not want to obey the rules of the babysitter.
Sebastian is not obeying Alex.

(6) STEFFI GRATULIERT MARTINA NICHT
Steffi congratulates Martina not

(a) Xaver veranstaltet eine große Geburtstagsparty. Steffi und Martina sind auch eingeladen, obwohl sie sich letztes Wochenende sehr mit Xaver gestritten haben. Steffi ist nicht nachtragend, aber Martina bleibt stur.
Xaver is hosting a big birthday party. Steffi and Martina are also invited although they had an argument with Xaver last weekend. Steffi does not hold a grudge, but Martina remains stubborn.
Steffi is congratulating, Martina is not.

(b) In der kleinen Dorfkapelle findet heute eine Hochzeit statt. Die Braut Martina hat auch ihre alte Schulfreundin Steffi eingeladen. Steffi ist jedoch schon lange Single und daher ziemlich verbittert.
In the little village chapel, a wedding is taking place today. Martina, the bride, also invited Steffi, her old friend from school. However, Steffi has been single for a long time and is therefore very bitter.
Steffi is not congratulating Martina.

(7) LISA WIDERSPRICHT BIANCA NICHT
Lisa contradicts Bianca not

(a) Herr und Frau Müller wollen, dass ihre Töchter Lisa und Bianca mehr im Haushalt mithelfen. Um die Aufgaben gerecht zu verteilen, hat Frau Müller einen Putzplan entworfen. Lisa hat keine Lust, sich an den Putzplan zu halten, während Bianca die Idee gut findet.
Mr. and Ms. Müller want their daughters Lisa and Bianca to help more with the housekeeping. In order to distribute the duties in a fair way, Ms. Müller has created a cleaning plan. Lisa does not want to stick to the cleaning plan while Bianca likes the idea.
Lisa contradicts, Bianca does not.

(b) In Toms Clique gibt's immer viel Zündstoff für Diskussionen. Besonders Lisa und Bianca liegen sich regelmäßig in den Haaren. Aber diesmal bleibt Lisa ganz ruhig als Bianca ihr wieder Vorwürfe macht.
In Tom's clique there are always lots of things to discuss. Lisa and Bianca in particular argue regularly. However, this time Lisa remains quiet when Bianca makes accusations against her again.
Lisa does not contradict Bianca.

(8) ANNE HEIRATET BARBARA NICHT
Anne marries Barbara not

(a) Letzte Woche fand in der Schule ein 10-jähriges Klassentreffen statt. Anne und Barbara haben sich lange nicht gesehen und fallen

sich freudig in die Arme. Während Anne stolz von ihrer anstehenden Hochzeit erzählt, denkt Barbara traurig an die Auflösung ihrer Verlobung.
A 10-year class reunion took place at the school last week. Anne and Barbara haven't seen each other for a long time and greet each other joyfully. While Anne is talking proudly about her upcoming wedding, Barbara is thinking sadly about the dissolution of her engagement.
Anne is marrying, Barbara is not.

(b) Die Gay-Community freut sich, dass gleichgeschlechtliche Ehen nun in den ganzen USA legalisiert wurden. Barbara und Anne sind schon lange ein Paar, daher stellt Barbara nun endlich die Frage aller Fragen. Aber Anne liebt Barbara nicht mehr und lehnt den Antrag ab.
The gay community is happy about the legalization of same-sex marriages in the USA. Barbara and Anne have been a couple for a few years. This is why Barbara finally pops the question. However, Anne does not love Barbara anymore and rejects the proposal.
Anne is not marrying Barbara.

(9) SUSI WÄSCHT ANDREA NICHT
Susi washes Andrea not

(a) Im Luise-Wohnheim gibt es einen großen Wäscheraum. Susi und Andrea treffen sich dort jeden Sonntagmorgen um Wäsche zu waschen. Eines Morgens verschläft Andrea aber leider, weil sie Samstag zu lange auf der Party war.
In the residential accommodation "Luise," there is a big laundry room. Every Sunday morning, Susi and Andrea meet there in order to do their laundry. Unfortunately, one morning Andrea oversleeps because she stayed at a party for too long.
Susi is washing, Andrea is not.

(b) Im Pflegeheim arbeiten viele freiwillige Helfer. Seit ein paar Wochen ist Susi für Bewohnerin Andrea zuständig. Susi darf aber bisher nur einfache Aufgaben übernehmen, wie z.B. beim Essen helfen.
A lot of volunteers work in the nursing home. Susi has been responsible for resident Andrea for a couple of weeks. So far, Susi is only allowed to do light tasks, like helping with meals, for instance.
Susi is not washing Andrea.

(10) ANTON BETRÜGT MARIA NICHT
Anton cheats (on) Maria not

(a) Die Geschwister Maria und Anton treffen sich regelmäßig zum Pokern in ihrer Lieblingskneipe. Oft wird dabei auch um Geld

gespielt. Während es Maria wichtig ist, fair zu spielen, versucht Anton immer zu tricksen.
The siblings Maria and Anton regularly meet each other to play poker at their favorite bar. They also often play for money. While it is important to Maria to play fairly, Anton always tries to cheat.
Anton cheats, Maria does not.

(b) Bei Maria und Anton läuft es schon länger nicht mehr so richtig in der Beziehung. Als sie ihn mit einer anderen Frau in einem Restaurant sieht, ist sie überzeugt, dass Alex eine Affäre hat. Im Nachhinein stellte sich aber heraus, dass es sich bei der Frau nur um seine Schwester handelte.
Maria and Anton's relationship is not what it used to be anymore. When she sees him with another woman at a restaurant, she is convinced that Alex is having an affair. Afterwards it turned out that the woman is his sister.
Anton does not cheat on Maria.

(11) ELIAS HILFT LUKAS NICHT
Elias helps Lukas not

(a) Auf dem Nachhauseweg werden Elias und Lukas Zeugen eines Zusammenstoßes zwischen einem Auto und einem Radfahrer. Der am Boden liegende Radfahrer schreit vor Schmerzen. Während Elias sofort losrennt, ist Lukas vor Schock wie gelähmt.
On their way home, Elias and Lukas witness a crash between a car and a bicycle. The cyclist on the ground is screaming in pain. While Elias immediately starts to run, Lukas is paralyzed with shock.
Elias is helping, Lukas is not.

(b) Um versetzt zu werden, muss Lukas mindestens eine 3 in Mathe schreiben. Verzweifelt richtet er sich an seinen älteren Bruder Elias. Dieser hat jedoch keine Zeit ihm Nachhilfe zu geben, da er sich lieber mit seiner neuen Freundin trifft.
Lukas has to get at least a C in math in order to go on to the next grade. He desperately looks for help from his brother Elias. However, Elias does not have time to help him because he prefers to meet with his new girlfriend.
Elias is not helping Lukas.

(12) TINE BERÄT OLGA NICHT
Tine advises Olga not

(a) Tine und Olga arbeiten beide für Mercedes, jedoch in unterschiedlichen Bereichen. Tine ist im Verkauf beschäftigt und hilft den Kunden das passende Auto zu finden. Olga hingegen arbeitet in der Produktion und überwacht dort die Arbeitsabläufe.

Tine and Olga work for Mercedes, but in different departments. Tine is working for the sales department and helps customers to find the right car. Olga, however, works in production and supervises the operations there.
Tine gives advice, Olga does not.

(b) Am Ende des Jahres will Olga ihre Steuererklärung machen. Da ihre Freundin Tine in einer Steuerberatungskanzlei arbeitet, bittet sie diese um Unterstützung. Jedoch hat sich Tine vorgenommen, Privates und Berufliches strikt zu trennen.
At the end of the year, Olga wants to file her tax return. Tine works for a tax consultancy office. This is why Olga asks for Tine's help. However, Tine wants to keep a strict separation between her private and professional affairs.
Tine does not give advice to Olga.

Notes

1 This chapter has benefited from valuable comments from the editors of this volume, Matthias Bauer and Angelika Zirker, as well as two anonymous reviewers. We are also grateful to the members of the Research Training Group (RTG 1808) and the audience of the "Fragments" workshop at the University of Saarbrücken in 2016 for their feedback and ideas. An additional thank you goes to Kirsten Brock for copyediting the paper. This material is based upon work supported by the German Research Foundation (DFG) under grant RTG 1808 (project number: 198647426).

2 Stripping, frequently also referred to as *bare argument ellipsis*, is defined by Hankamer and Sag (409) as "a rule that deletes everything in a clause under identity with corresponding parts of a preceding clause, except for one constituent." See also Konietzko for different types of stripping constructions.

3 Tones and Break Indices (ToBI) is a phonological intonation scheme originally developed for English (Silverman et al.).

4 We take trouble detection to subsume the detection of errors and other challenges like ambiguities; likewise, repair is not restricted to errors.

5 Both factors are within items factors in the F_2 analysis; see the section "Analysis of Prosodic Phrasing".

6 We are aware that a third reading, namely, an OVS structure, is possible. The sentence would thus be understood as *Nadine isn't bathing Janina*. Because this reading is highly dispreferred and not suggested by any of our contexts, we disregard the OVS reading.

7 We wanted to ensure that the participants would read both context versions rather than only the highlighted one. Therefore, we asked participants of the Inf-Group to tell us what they were currently doing. That is, while reading context one, participants said something like, "Now, I'm reading context one" or "Text one."

8 "[I]nformation-structurally given material is subject to prosodic reduction (*givenness marking hypothesis*) and material that remains must be prosodically highlighted (*contrastive remnant condition*)" (Winkler, "Ellipsis and Prosody" 360; emphasis in original).

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